Fiscal Year:	FY 2023	**Task Last Updated:**	FY 09/24/2022
PI Name:	Jung, Sunghwan Ph.D.		
Project Title:	Enhanced Particle Tracking in STDC Data and Its Industrial Application of Marangoni Convection		
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Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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PI Email:	sj737@cornell.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	607-255-5798
Organization Name:	Cornell University		
PI Address 1:	Department of Biomedical and Environmental Engineering		
PI Address 2:	111 Wing Drive		
PI Web Page:			
City:	Ithaca	**State:**	NY
Zip Code:	14853	**Congressional District:**	23
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Contact Monitor:	Hasan, Mohammad	**Contact Phone:**	216-977-7494
Contact Email:	Mohammad.M.Hasan@nasa.gov		
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The Principal Investigator (PI) proposes to extract/understand detailed thermo-fluid mechanics of previous experiments conducted by NASA's Physical Sciences Informatics (PSI) STDCE-1 (Surface Tension Driven Convection Experiment). STDCE provided invaluable video recordings (qualitative data), but only a few papers/reports were published with an outdated particle tracking method in early 1990. Since then, there have been a number of particle tracking methods developed with various filtering functions, sub-pixel identification and more, which would allow us to access more information from in-space STDC experiments. Moreover, our proposed ground-based STDC experiments including gravity effect can allow us to identify/separate STDC from gravitational convection, which was not possible before. The overall goal of the proposal is to advance our understanding of thermo-fluid mechanics in surface tension driven convection through state-of-the-art particle tracking methods and ground-based experiments, which would be beneficial

Task Description:	for industrial STDC-based drying or cleaning processes. Our proposed research activities are classified into three major tasks: Task 1-Particle tracking analysis: Particle tracking analysis was performed in 1994 by the original designer of the Surface Tension Driven Convection Experiment (STDCE). Since then, many other advanced particle tracking methods have been developed. In this task, the PI's group will compute and compare velocity fields of the PSI STDCE videos using several up-to-date particle tracking methods (four MATLAB-based codes and three ImageJ-based codes). Furthermore, pressure fields will be calculated using multi-point integration methods. Therefore, this task will provide fluid velocity and pressure fields, which will be compared for later tasks. Task 2-Theoretical and computational models: In this task, STDC flow and pressure fields will be theoretically computed using the perturbation method. Also, computational solutions from COMSOL will be obtained with appropriate boundary conditions including a curved free surface. These theoretical and computational flow and pressure fields will serve as a basis to validate the results of various particle tracking methods in Task 1. Task 3-Ground-based experiment for industrial processes: The industrial manufacturing sector has utilized surface tension driven convection for drying and cleaning processes, which is critical to enhance nano-fabrication processes of silicon integrated circuit (IC) chips or liquid crystal displays. In this task, the PI proposes ground-based experiments to understand the STDC mechanism with a curved top free surface similar to in-space STDCE. By comparing ground-based STDC experiments with in-space STDCE results, we can distinguish the effect of gravitational convection from pure surface tension driven convection. Relevance to NASA-PSI: NASA's Physical Sciences Informatics (PSI) obtained a large amount of invaluable data from previous space experiments on the International Space Station. However, most of the data are limited to only video recordings and analyses were done quite a long time ago. While the video recordings captured accurate 2D motions of seeding particles in the STDC systems, more thermo-fluidic information can be extracted through state-of-the-art particle tracking, sub-pixel particle localization, and pressure field calculation. Our proposed activities could revitalize the PSI STDCE data sets, whose knowledge could be translated into a better understanding of ground-based industrial drying and cleaning processes of integrated circuits and liquid crystal displays.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	By revisiting the NASA STDCE experiments, we have broadly developed three key understandings about marangoni-driven interfacial flow problems which would be crucial for exploiting marangoni effect for cleaning surfaces. First, we have identified the best diagnostic tool to analyze the flow fields in PIV/PTV experiments. Second, our numerical model identifies the key geometric and material parameters and their combination in dimensionless groups, which dictate the response of a marangoni driven flow system. Third, we identified the role of a curved interface in governing the flow structure. On the other hand, one of the leading motivations for research on synthetic active matter systems is to develop novel medical diagnostic tools (micro/nano robots). Our study, as part of the second year's work, sheds light on the importance of the bulk flow and vortex structures below such a camphor swimmer which is relevant under shallow water level conditions, an expected condition for micro/nano robots for targeted drug delivery etc. Our experiments on the interaction with passive surface particles will shed light on how surface active swimmers can be used to control contaminants on an interface and potentially serve as motivation for a novel surface cleaning technique or microplastics removal technique.
Task Progress:	The Marangoni effect, flow towards a high surface tension region from a low surface tension region at an interface, plays an important role in many industrial applications where a surface tension gradient exists. For instance, the cleaning process of silicone wafers in integrated circuits incorporates this effect to remove thin layers of water from the surface by applying a chemical gradient. Interfacial active swimmers are specific types of synthetic active matter systems that utilize the Marangoni effect for their self-propulsion at the air-water interface. As mentioned in the planned work section of the previous year's report, this year we have studied the bulk flow characteristics of interfacial camphor swimmers and their interactions with a layer of passive particles at the air-water interface. Although the interfacial camphor swimmer system has been studied for many years, the bulk flow below the swimmer has not been investigated in detail. Here we have experimentally visualized the bulk flow field below a freely moving camphor swimmer using Particle Image Velocimetry (PIV). From the experimental flow field, we observed a stable vortex structure just below the swimmer as it self-propels. Interestingly, the structure of this vortex changes with time as the swimmer performs a turning motion. Being able to understand the role of the vortex structure on self-propulsion will help us predict how the swimmer motion is affected with varying water levels. Second, with a goal of understanding how such synthetic active swimmers behave in complex environments, we have also looked at how these swimmers interact with different packing fractions of passive particles on the interface. We have observed a clear behavior change in the dynamics of the swimmers above a critical initial packing fraction. This study will enlighten us about how contaminants can affect the motion of such self-propelling systems.
Bibliography Type:	Description: (Last Updated: 11/28/2023)
Abstracts for Journals and Proceedings	Maitra A, Rotheram J, Pandey A, Jung S. "Transient behavior of a Marangoni swimmer." 75th Annual Meeting of the APS Division of Fluid Dynamics, Indianapolis, Indiana, Nov. 20-22, 2022. Bulletin of the American Physical Society. 2022 Nov:Z04.00009. https://meetings.aps.org/Meeting/DFD22/Session/Z04.9 , Nov-2022
Articles in Peer-reviewed Journals	Sequeira Y, Maitra A, Pandey A, Jung S. "Revisiting the NASA surface tension driven convection experiments." npj Microgravity. 2022 Feb 18;8(5). https://doi.org/10.1038/s41526-022-00189-5 , Feb-2022